Reservoir Quality Assessment Using Wireline Log Data

1. Introduction

This project presents a petrophysical evaluation of a subsurface reservoir using real well log data (LAS format). The objective is to estimate key reservoir properties including shale volume, porosity, water saturation, and net pay zones using Python-based analysis.

---

2. Data Description

The dataset consists of wireline logs obtained from a field well, including:

- Gamma Ray (GAMMA)
- Density Log (DEN(CDL))
- Density Porosity (POR(DEN))
- Resistivity Logs (RES(DI), RES(MI), RES(GRD))
- Caliper and SP logs

These logs were processed and analyzed using Python libraries such as NumPy, Pandas, and Matplotlib.

---

3. Methodology

3.1 Data Cleaning

Missing values (-999.25) were replaced with NaN and filtered to ensure data quality.

3.2 Volume of Shale (Vsh)

Calculated using the Gamma Ray index:

$$V_{sh} = (GR - GR_{min}) / (GR_{max} - GR_{min})$$

3.3 Porosity

Density porosity (POR(DEN)) was used directly as total porosity.

3.4 Water Saturation (Sw)

Calculated using the Archie equation:

$Sw = ((a * Rw) / (\varphi^m * Rt))^{(1/n)}$

Assumptions:

- $a = 1$
- $m = 2$
- $n = 2$
- $Rw = 0.1$

3.5 Net Reservoir and Net Pay

Reservoir criteria:

- Vsh < 0.35

Pay criteria:

- Vsh < 0.35
- Porosity > 0.10
- Sw < 0.60

---

4. Results and Interpretation

4.1 Shale Volume

Low gamma ray zones indicate cleaner sand intervals with low shale content.

4.2 Porosity

Porosity values generally range between 0.10–0.30, indicating moderate to good reservoir quality.

4.3 Water Saturation

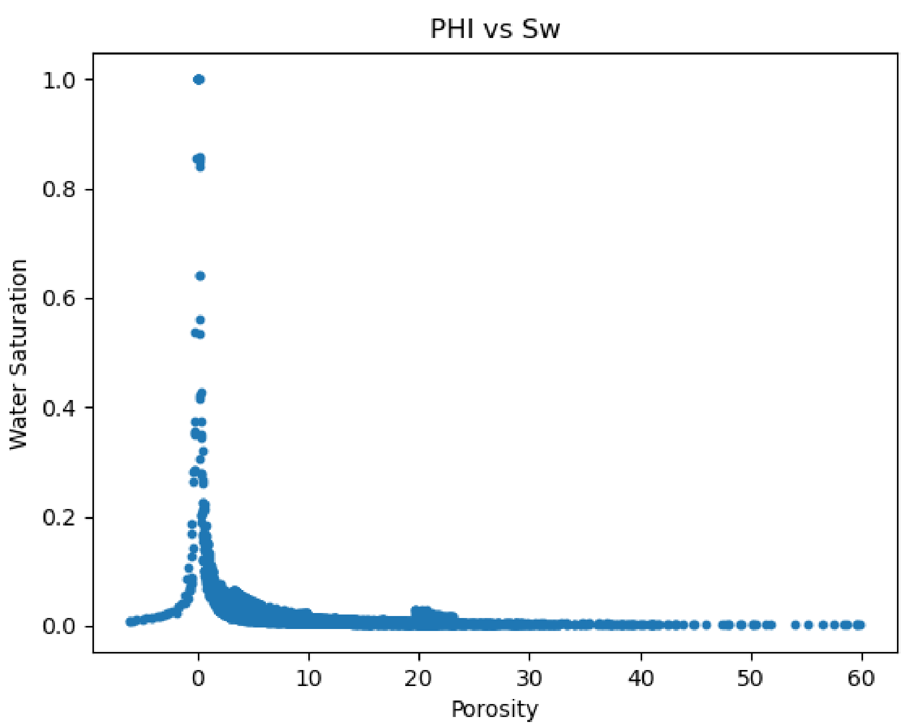
Lower Sw zones (<0.6) suggest potential hydrocarbon-bearing intervals.

4.4 Reservoir Zones

Intervals with low Vsh and sufficient porosity were identified as potential reservoir zones.

4.5 Net Pay Zones

Zones satisfying all criteria (low Vsh, high porosity, low Sw) were classified as net pay and highlighted in the well log plots.



5. Visualization

A multi-track well log plot was generated including:

- Gamma Ray
- Vsh
- Porosity
- Water Saturation
- Pay zone highlighting

---

6. Limitations

- Archie parameters (a, m, n) were assumed due to lack of core data
- No neutron log available for porosity cross-validation
- Shale effects not explicitly corrected

---

7. Conclusion

The analysis successfully identified potential reservoir and pay zones using real well log data. The workflow demonstrates how Python can be effectively used for petrophysical evaluation, providing a cost-effective alternative to commercial software.

---

8. Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- LASIO

---

9. Key Skills Demonstrated

- Well log interpretation
- Petrophysical analysis
- Data cleaning and processing
- Reservoir evaluation
- Scientific programming in Python.